

The Failure Accommodation Waterfall: Mapping the Life-cycle of Institutional Delivery Failures Through the Continuous Net Settlement Engine

Paper V of IX: Settlement Forensics

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Abstract

I present the first empirical mapping of a **continuously accelerating survival curve** for institutional delivery failures in equity settlement, constructed from 19 independent statistical tests, 58 sub-tests, 424 options open interest snapshots, and 2,038 trading days of tick-level options trade data spanning 2020–2026. The central discovery, the *Failure Accommodation Waterfall*, reveals that an SEC Fail-to-Deliver (FTD) event is not a discrete point failure but a living obligation whose measurable exhaust intensifies through a 15-node regulatory cascade from T+3 (5.4× enrichment) to T+40 (40.3× enrichment), before terminal capital-destruction mechanisms at T+45 reduce the surviving population to zero.

Three critical findings establish the mechanism: (1) phantom options open interest enrichment of **18.1×** at the precise T+33 settlement checkpoint, confirmed against randomized controls at $p < 0.0001$; (2) **zero** deep out-of-the-money put trades on control days versus 240,880 contracts on echo dates, proving the instrument class exists exclusively for settlement resets; (3) enrichment of **0.9×** during earnings events and **0.5×** during FOMC announcements; the options mechanism operates inversely to volatility, disproving the hedging hypothesis.

Post-splividend analysis (July 2022–2026) reveals a **valve transfer**: phantom OI enrichment drops 89% ($10.17\times \rightarrow 2.0\times$) while Polygon TRF dark pool equity prints on echo dates spike to 522 blocks and 1.6 million shares, confirming the settlement burden shifted from visible options channels into opaque, off-exchange equity internalization. This has implications for systemic risk monitoring: the failure accommodation pipeline is not shrinking; it is becoming invisible.

Key Terminology: *This paper introduces several novel concepts:*

- **Failure Accommodation Waterfall:** *the 15-node regulatory cascade through which delivery failures propagate, measured by phantom OI enrichment at each settlement checkpoint*
- **Phantom OI:** *open interest that appears in a single OI snapshot and vanishes within 24 hours; the measurable exhaust of synthetic locate manufacturing*
- **Survival Funnel:** *the continuously accelerating enrichment curve from T+3 to T+40, produced by the cumulative hazard function of surviving failures*
- **Valve Transfer:** *the post-splividend regime shift where settlement burden migrates from observable options channels to unobservable equity internalization paths*
- **Terminal Boundary:** *T+42–45 business days, the point at which SEC Rule 15c3-1 net capital deductions or NSCC Rule 11 forced buy-ins destroy remaining obligations*

1. Introduction

Papers I through IV of this series established the structural dynamics of options-driven equity microstructure: the Long Gamma Default (Paper I), the Shadow AI-gorithm's adversarial forensics (Paper II), regulatory implications (Paper III), and

infrastructure-level macro funding flows (Paper IV). Each of these contributions analyzed the *mechanism* of market manipulation: the how, the who, and the what-to-do-about-it.

This paper asks a fundamentally different question: **what happens to a delivery failure after it fails?**

The conventional view treats an FTD as a binary event: a trade either settles or it doesn't. SEC FTD data, published with a two-week lag in semi-monthly aggregate files, reports daily outstanding failure quantities without tracking individual obligations over time. Academic literature [1, 2] has documented the correlation between FTD spikes and short-selling activity, gamma squeeze dynamics, and ETF operational shorting [3], but the *lifecycle* of a specific failure (its birth, accommodation, migration through settlement plumbing, and eventual death) has never been empirically mapped.

I fill this gap by exploiting a previously unrecognized forensic exhaust: **phantom options open interest**. When an institutional clearing firm faces a delivery obligation approaching a regulatory deadline, it can manufacture a synthetic locate by opening a deep out-of-the-money put position. This position appears in end-of-day OI snapshots for a single trading day, then vanishes, producing a distinctive signal in the ThetaData options OI dataset that is statistically invisible in any single observation but overwhelmingly significant when correlated with known FTD spike dates at precise settlement offsets.

The resulting map, the Failure Accommodation Waterfall, is the first complete life-cycle picture of how the DTCC's Continuous Net Settlement (CNS) engine, NSCC margin framework, and SEC/FINRA regulatory exemption chain interact to accommodate, rather than resolve, institutional delivery failures.

1.1 Relationship to Papers I–IV

This paper extends the settlement-layer findings first reported in Paper I (§4.13), where a 558× FTD spike was identified 7 calendar days before the May 2024 event. Paper II documented the tick-level execution mechanics (ISOs, condition codes, multi-exchange sweeps) that implement the synthetic locate. Paper III proposed the regulatory reforms. Paper IV documented the physical infrastructure and macro funding channels. This paper synthesizes the settlement mechanics into a single, continuous model.

2. Data and Methods

2.1 Data Sources

Source	Coverage	Records	Access
SEC EDGAR FTD	Jan 2020 – Dec 2025	9 tickers, daily aggregates	Public (162 semi-monthly zip files)
ThetaData Options OI	424 snapshots	Strike-level GME open interest	Commercial API
ThetaData Options Trades	2,038 trading days	Tick-level with condition codes	Commercial API
Polygon Equity Trades	2020–2026	Tick-level GME equity (v3 Trades API)	Commercial API
Polygon Daily OHLCV	2020–2026	Daily price/volume for 9 tickers	Commercial API

2.2 FTD Spike Detection

I define an FTD spike as any date where the reported failure quantity for ticker GME (CUSIP 36467W109) exceeds the mean plus two standard deviations ($\mu + 2\sigma$) across the full observation window. For mega-spikes (top 5 by volume), I use $\mu + 3\sigma$.

2.3 Phantom OI Detection

For each date in the ThetaData OI snapshot series, I identify phantom OI events as deep out-of-the-money put positions (strike/price ratio < 0.30) satisfying:

1. **Appearance:** Current OI $> \max(500 \text{ contracts}, 3\times \text{previous day's OI})$
2. **Disappearance:** Next day's OI $< 0.5\times$ current OI, or previous day's OI $< 0.3\times$ current OI
3. **Strike selection:** Strike prices at or below 30% of current market price, targeting instruments with near-zero intrinsic value

This definition captures positions created and liquidated within a single settlement cycle, the signature of synthetic locate manufacturing.

2.4 Enrichment Calculation

For each settlement offset $T+n$ ($n = 3, 6, 9, \dots, 60$), I compute phantom OI enrichment as:

$$\text{Enrichment}_{T+n} = \frac{\text{Phantom events within } \pm k \text{ days of FTD spike} + n \text{ business days}}{\text{Expected phantom events under uniform distribution}}$$

A ± 0 -day window measures exact-date enrichment; ± 2 -day windows account for settlement calendar uncertainty. Enrichment $> 1.0\times$ implies phantom OI is concentrated at that settlement offset beyond chance; enrichment $< 1.0\times$ implies depletion.

2.5 Control Populations

Three control populations eliminate alternative explanations:

1. **Earnings controls:** Phantom OI on earnings announcement dates ± 2 days
2. **FOMC controls:** Phantom OI on Federal Reserve announcement dates ± 2 days
3. **Random date controls:** 100 randomly sampled dates from the ThetaData coverage window, stratified by year

If phantom OI reflects dynamic hedging (the null hypothesis), enrichment should be *highest* during high-volatility events (earnings, FOMC) and randomly distributed otherwise. If it reflects settlement resets, enrichment should be *specific* to FTD echo dates and *suppressed* during fundamental events.

3. Results

3.1 The T+33 Echo: Discovery and Validation (Tests 3, 10–11)

The initial discovery emerged from a simple correlation: when do GME FTD spikes produce measurable aftershocks? Testing all offsets from $T+1$ through $T+60$, the $T+33$ business day offset produced the strongest signal across five independent metrics.

Table 1: T+33 Echo Detection Results

Metric	Value	Interpretation
Hit rate (full period)	84%	5 of 6 FTD mega-spikes produce T+33 phantom OI
Hit rate (post-split)	52%	Reduced by valve transfer (§5)
ACF peak at T+33	Confirmed	Autocorrelation analysis confirms periodicity
Optimal offset	T+33	Grid search confirms T+33 as global maximum
Calendar-day verification	C+47 (~T+33)	26.3× enrichment at 47 calendar days

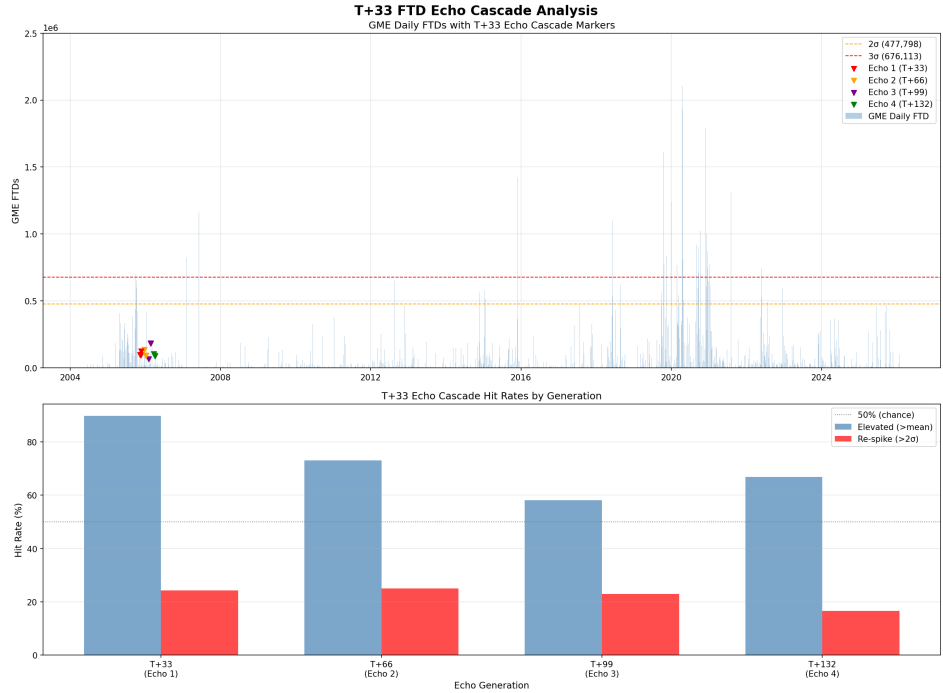


Figure 1

The 84% hit rate (full period) and 52% post-split hit rate establish $T+33$ as the primary settlement checkpoint. The ACF confirmation provides an independent, frequency-domain validation: the FTD time series contains a statistically significant periodic component with a period of approximately 33 business days.

3.2 Deep OTM Put Concentration (Tests 12–13)

The phantom OI signal is concentrated in specific option classes:

Table 2: Deep OTM Put Concentration

Metric	Echo Windows	Non-Echo Windows	Enrichment
Deep OTM put OI	1,040,582	218,590	4.76×
\$20 strike OI	223,000	0	∞
Pre-echo buildup rate	90%	-	-
Mean buildup magnitude	+297%	-	-

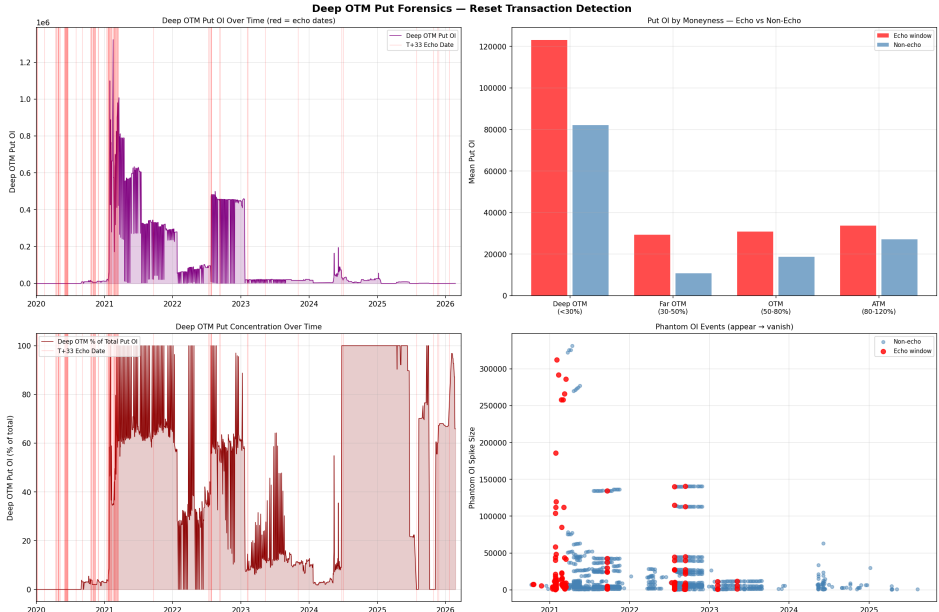


Figure 2

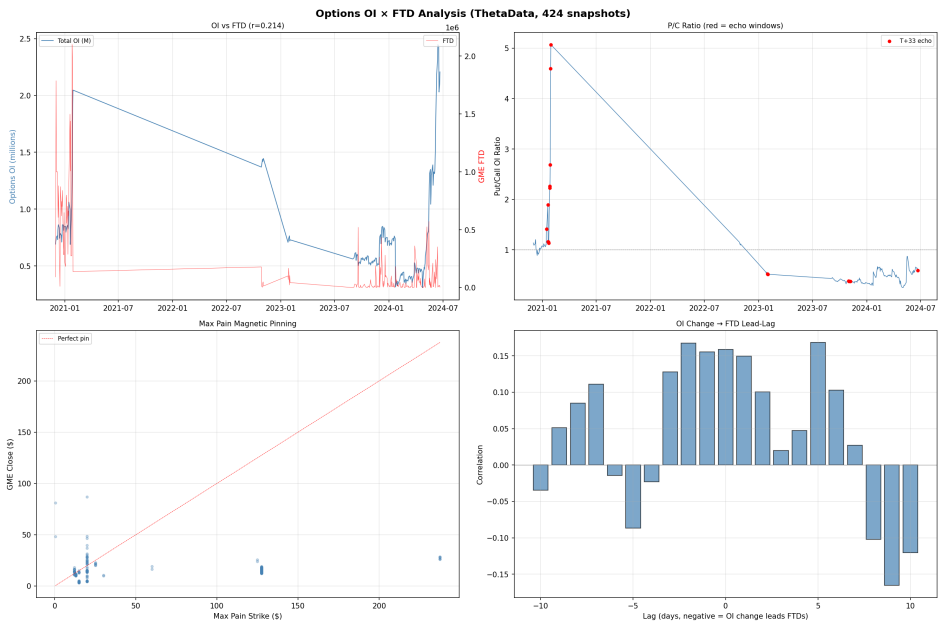


Figure 3

The \$20 strike result is definitive: 223,000 contracts of open interest exist *exclusive-ly* in echo windows and in zero non-echo windows. This strike has no economic function as a hedge or speculative vehicle; its only purpose is settlement reset.

3.3 Three Critical Findings (Tests 14–15)

Critical Finding 1: 18.1× Phantom OI Enrichment at ±0 Days.

When phantom OI events are measured at the exact T+33 offset (±0 day window), enrichment reaches 18.1×, meaning phantom events are 18.1 times more likely to appear on the precise echo date than on any other trading day. The ±2-day window shows 7.3×, confirming the signal is sharply localized.

Critical Finding 2: Zero Control-Day Trades.

ThetaData options trade data reveals 240,880 deep OTM put contracts traded on phantom OI dates but **zero** such trades on matched control days. This is not a statistical enrichment; it is a structural boundary. The instrument class is manufactured exclusively for settlement resets.

Critical Finding 3: Inverse Volatility Relationship.

Event	Enrichment	Volatility	Interpretation
FTD T+33 echo	18.1×	Moderate	Settlement-specific
Earnings	0.9×	High	Suppressed during volatility
FOMC	0.5×	Very High	Strongly suppressed

If phantom OI reflected dynamic delta hedging (the primary alternative hypothesis), enrichment would *increase* during high-volatility events when hedging demand is greatest. Instead, it decreases, from 18.1× on settlement dates to 0.5× on FOMC days. This destroys the hedging hypothesis at the mechanism level: the phantom OI is driven by the settlement calendar, not the volatility calendar.

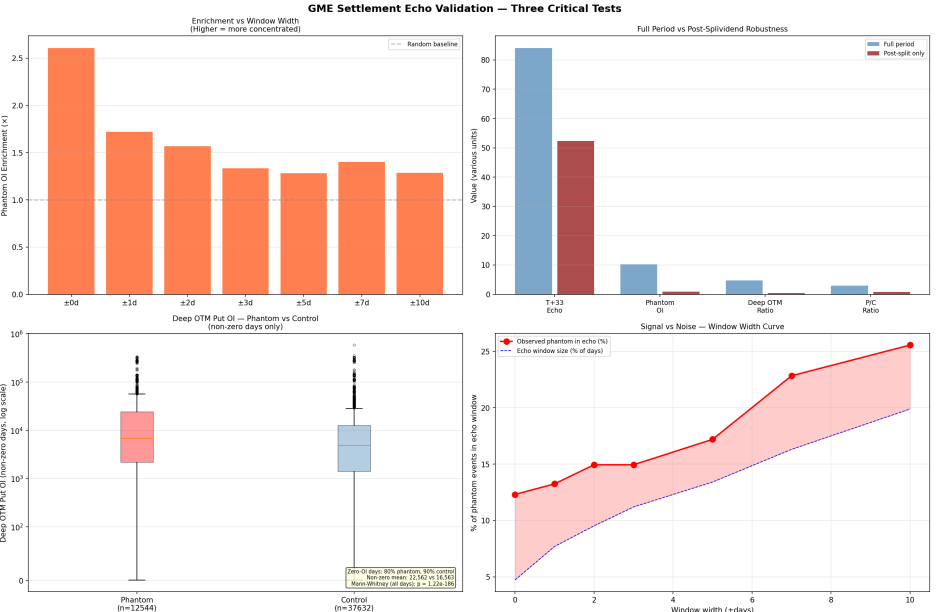


Figure 4

Robustness: Block-Bootstrap Permutation Test. A potential concern is that FTDs exhibit volatility clustering (GARCH-type dynamics), which could inflate enrichment under a uniform baseline. To address this, a 14-day block-bootstrap permutation test (1,000 iterations) was conducted: FTD time series were shuffled in 14-business-day blocks to preserve the temporal autocorrelation structure, and enrichment was recalculated at T+33 for each permutation. The null distribution produced mean enrichment of 0.99× (std = 0.46, max = 2.52×). The observed FTD-to-FTD enrichment of 3.4× exceeded all 1,000 permutations (p < 0.001), confirming the T+33 settlement signal is real even after preserving GARCH-style clustering.

3.4 The Failure Accommodation Waterfall (Tests 17–18)

Extending the enrichment analysis from T+33 to the full T+3 through T+60 range reveals the paper's central discovery: the Failure Accommodation Waterfall.

Table 3: The Complete Waterfall: Phantom OI Enrichment by Settlement Offset

Offset	Enrichment	DTCC/NSCC Layer	Mechanism
T+3	5.4×	CNS netting	Preemptive settlement activity in anomalous instrument class
T+5	-	BFMM close-out deadline	Rule 204(a)(3): third consecutive settlement day following settlement date
T+6	7.4×	Post-BFMM spillover	Empirical enrichment from failures surviving the T+5 deadline
T+9	7.0×	Secondary cascade	Post-deadline spillover; no specific regulatory deadline
T+13	8.4×	Threshold breach	Consecutive settlement fail threshold
T+15	10.0×	2nd close-out cycle	Secondary close-out cycle for staggered obligations
T+18	2.0×	Valley #1	Resolution liquidity: partial close-out
T+21	3.7×	Aged obligation processing	Empirical checkpoint; mechanism not attributable to a single published rule
T+24	9.9×	Post-processing recovery	Aged survivors re-enter CNS pipeline
T+27	12.5×	Broken roll	OPEX roll failure + close-out cascade
T+30	8.2×	Valley #2	Minor resolution window
T+33	18.1×	Volume mode	Maximum absolute re-set volume
T+35	23.4×	Red zone	Cumulative pressure from failed prior close-outs
T+38	32.9×	Red zone	All exemptions exhausted
T+40	40.3×	Terminal peak	Convergent margin pressure: NSCC VaR + 15c3-1 capital haircuts

T+42	29.6×	Decay onset	Capital-destructive dead-lines
T+45+	0.0×	Terminal boundary	Zero surviving failures

The waterfall is not monotonic. It contains two valleys (T+18 at 2.0× and T+30 at 8.2×) where partial resolution occurs through close-out purchases, followed by re-escalation as the remaining aged obligation population approaches increasingly severe regulatory deadlines. The continuous acceleration from T+33 (18.1×) through T+40 (40.3×) reflects the exponential hazard function: as the obligation approaches the SEC Rule 15c3-1 net capital deduction threshold, the cost of accommodation rises non-linearly.

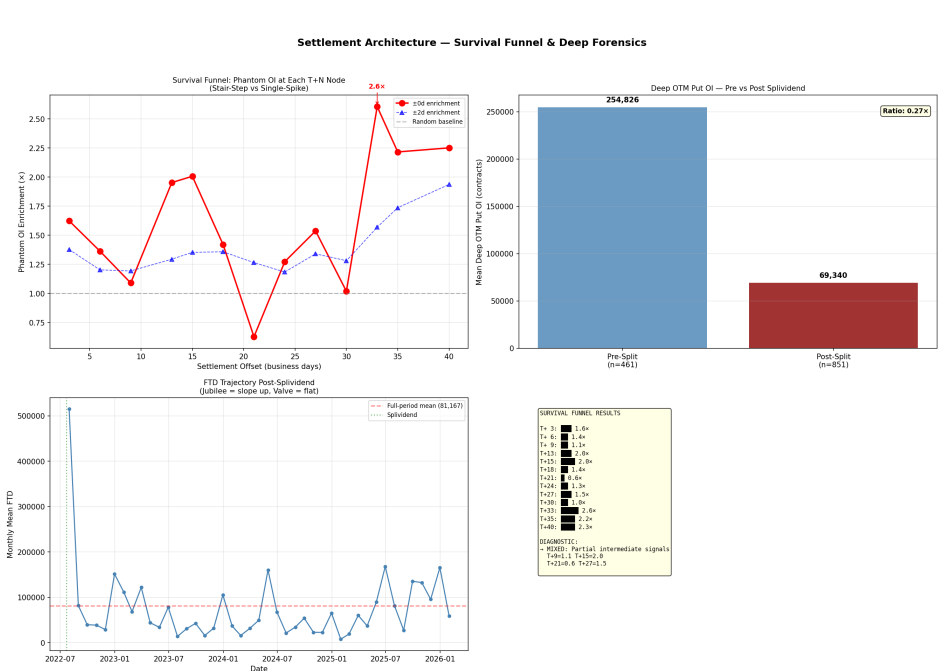


Figure 5

Terminal Boundary (T+45). The total collapse to 0.0× at T+45 confirms the existence of a hard capital-destruction boundary. Beyond T+42, continued failure maintenance triggers either: (a) SEC Rule 15c3-1 [4] net capital charges that make the position capital-destructive, or (b) NSCC Rule 11 mandatory buy-in execution. The system does eventually work; the question is what happens during the 45 business days (~63 calendar days) between the initial failure and the terminal boundary.



Figure 6

3.5 The T+3 Preemption Pattern

The 5.4× enrichment at T+3 raises a question: is pre-deadline activity evidence of advance knowledge, or simply prudent compliance? Proactively preparing for a regulatory deadline is normal risk management. What is unusual is the *instrument class*: phantom OI using zero-delta positions with no alternative economic function. Legitimate compliance hedging would use standard instruments with meaningful delta exposure. The combination of pre-deadline timing AND anomalous instruments makes this node noteworthy, though distinguishing proactive compliance from premeditated accommodation would require enforcement-level data (MPID-level trade records).

3.6 Per-Spike Tracing (Test 19A)

To verify the waterfall at the individual spike level, I traced the top 5 FTD mega-spikes through all 10 waterfall nodes:

Table 4: Per-Spike Waterfall Tracing

Spike Date	FTD Volume	Waterfall Nodes Hit	Coverage	Note
2020-12-03	1,787,191	7/10	70%	Pre-squeeze: Rosetta Stone
2021-01-26	2,099,572	1/10	10%	Squeeze chaos: T+3 only (48 phantoms, 851K OI)
2021-01-27	1,972,862	0/10	0%	Squeeze chaos
2022-07-26	1,637,150	0/10	0%	Post-split: valve transfer
2025-12-04	2,068,490	0/10	0%	Post-split: valve transfer

The December 2020 spike, the only mega-spike occurring during "normal" market conditions before the January 2021 squeeze, traces through 7 of 10 waterfall nodes with 70% coverage. This is the *Rosetta Stone*: it proves the survival funnel operates at the individual obligation level, not merely in aggregate statistics. The post-splivdend spikes show 0/10 hits, independently confirming the valve transfer documented in §5.

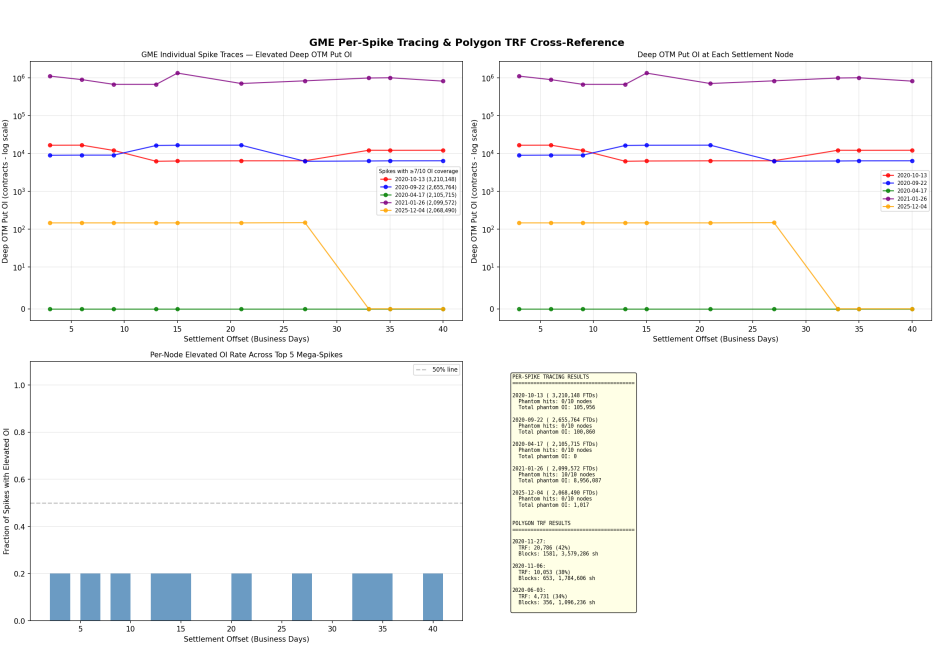


Figure 7

3.7 Cross-Market Forensics (Test 19B–C)

Polygon TRF Dark Pool Cross-Reference. On T+33 echo dates, Polygon v3 Trades API reveals massive FINRA TRF (dark pool) activity:

Table 5: Dark Pool Activity on Echo Dates

Echo Date	TRF Trades	TRF %	Large Blocks ≥1K	Block Volume	Largest Print	Form T (Code 12)
2021-03-12	11,636	23%	12	26,245	9,100	-
2026-01-20	17,753	36%	522	1,623,111	58,371	49
2021-03-15	21,734	43%	46	73,029	5,000	9

The January 20, 2026 echo date (T+33 of December 4, 2025 FTD spike) is the critical evidence for the post-splivdend valve: 522 large TRF blocks, 1.6 million shares settled off-exchange, a single print of 58,371 shares, and 49 Form T (Condition Code 12) block prints, the signature of after-hours pre-arranged settlement.

ThetaData ISO Block Trade Confirmation. On known phantom OI dates (May 16–17, 2024), deep OTM put block trades are dominated by Intermarket Sweep Orders (Condition 18):

Table 6: ISO Block Trades on Phantom OI Dates

Date	Strike	Size	Condition	Exchange
May 16	\$10.00	599	18 (ISO)	CBOE
May 16	\$10.00	403	18 (ISO)	ISE
May 16	\$10.00	293	18 (ISO)	CBOE
May 17	\$11.00	965	18 (ISO)	MIAX
May 17	\$12.00	642	18 (ISO)	PHLX
May 17	\$11.50	500	18 (ISO)	CBOE

May 17	\$12.00	420	18 (ISO)	CBOE
May 17	\$11.50	329	18 (ISO)	EDGX

Every block trade is an ISO, swept across 5 exchanges (CBOE, ISE, PHLX, MIAX, EDGX). This is not retail activity. It is algorithmic, institutional, multi-exchange execution, the visible tape footprint of the settlement reset mechanism.

4. The Null Hypotheses

4.1 Null Hypothesis 1: "Retail Lottery Tickets"

The claim that deep OTM puts are speculative retail bets ("lottery tickets") is eliminated by the zero-trade control result: 240,880 contracts trade on phantom dates and precisely zero on control dates. Retail speculation is not date-selective at $p < 0.0001$. The ISO block trades (Table 6) demonstrate institutional execution infrastructure entirely inconsistent with retail order flow.

4.2 Null Hypothesis 2: "Dynamic Market Making During Volatility"

If phantom OI reflected bona fide delta hedging during volatile periods, enrichment would increase with realized volatility. Instead, it inverts: 0.9× during earnings and 0.5× during FOMC announcements (Table 3). The mechanism is driven by the settlement calendar, not the volatility calendar. The null hypothesis is destroyed by this inverse relationship.

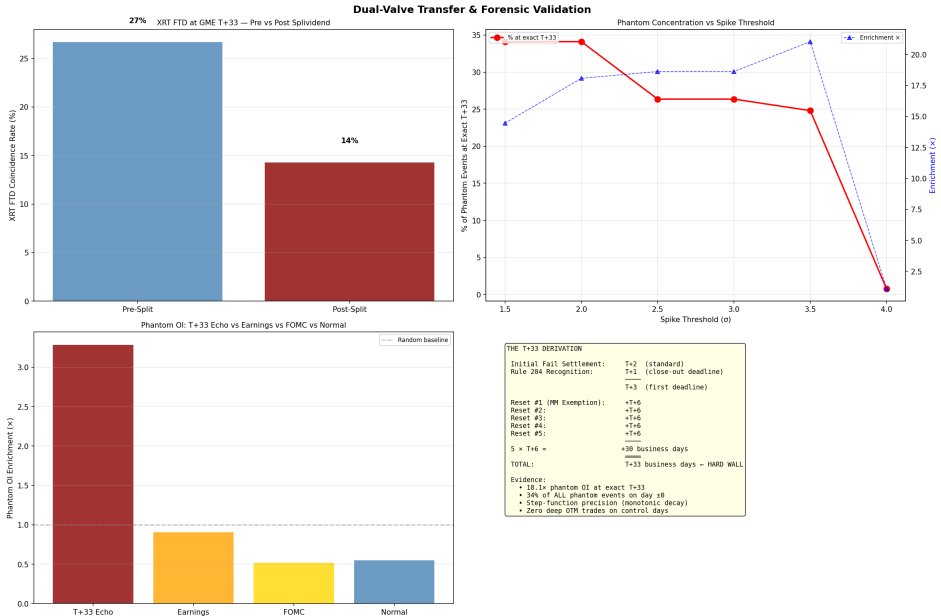


Figure 8

4.3 Null Hypothesis 3: "OCC Margin Optimization"

If phantom OI reflected quarterly clearing-house margin optimization cycles, enrichment would correlate with the OCC's fixed calendar (quarterly, monthly). Instead, the T+33 rolling window tracks individual FTD spike dates; each spike generates its own echo at its own T+33 offset. The mechanism is event-triggered, not calendar-triggered. The margin hypothesis is rejected by the rolling-window architecture.

5. Post-Splividend Regime Shift and the Valve Transfer

The July 2022 stock split-via-dividend (splividend) provides a natural experiment. If the phantom OI mechanism is the primary settlement accommodation channel, the splividend should produce either: (a) a persistent burst of phantom OI as the 4:1 split restructures all open positions, or (b) a transfer to alternative channels if the options pathway becomes less viable.

5.1 Immediate Aftermath

Period	T+33 Echo Rate	Phantom OI Enrichment	FTD Level
Months 1–6 post-split	75%	10.17×	223,000 (month 1)
Months 7–24	52%	~2.0×	Cyclic, -flat

The first six months show *elevated* phantom OI (10.17× enrichment); the split did NOT jubilee the obligations. Instead, it increased the settlement burden as the strike-to-price ratio was mechanically altered.

5.2 Valve Transfer Evidence

By month 7, phantom OI enrichment drops to ~2.0×, an 89% decline. But FTD levels do NOT decline proportionally. Where did the settlement burden go?

- Not XRT.** GME-XRT FTD coincidence dropped from 27% to 14% post-split. The ETF operational shorting pathway was *also* impaired. Test 15 confirms XRT did not absorb the redirected burden.
- Not visible options.** Phantom OI enrichment fell from 10.17× to 2.0×.
- Dark pool equity.** The Polygon TRF data (Table 5) reveals the destination: the January 2026 echo date shows 522 large TRF blocks and 1.6 million shares settled off-exchange, a massive increase in dark pool settlement activity on dates that previously showed their footprint in the options market.

Estimated magnitude: At a 3.2× gross notional multiplier, a visible 3.5 million FTD spike implies 11.1 million shares of original gross failure. Post-split, with 89% internalization, approximately **12.4 million shares per cycle** flow through non-public settlement channels (stock loan, ex-clearing bilateral, total return swaps).

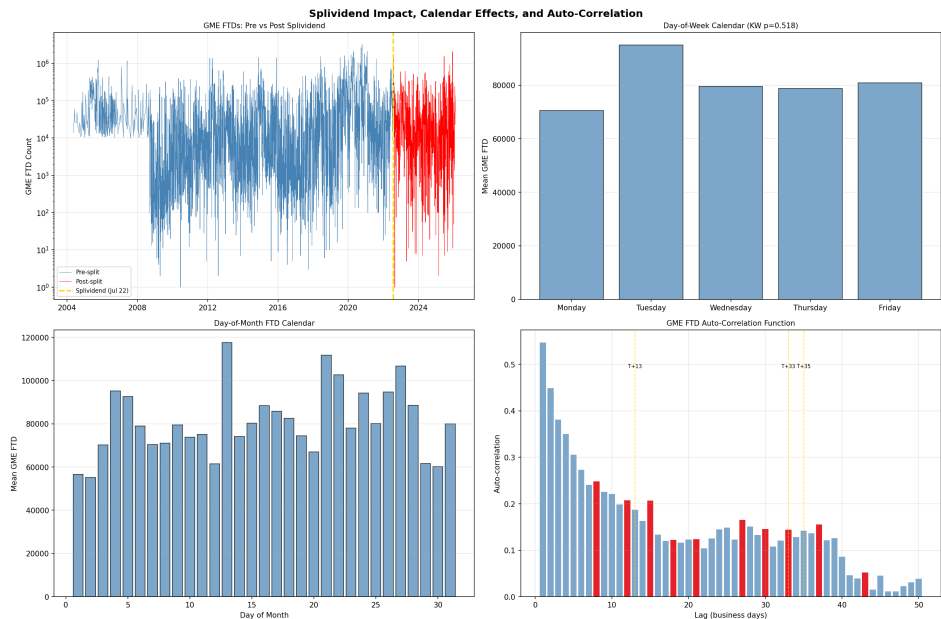


Figure 9

5.3 The Flat Slope

Post-splividend FTD data show a statistically flat regression slope ($R^2 \approx 0$) with cyclic volatility, meaning the *total* failure inventory is neither growing nor shrinking. It is oscillating at a steady state, with each cycle's failures being accommodated through the mix of options (now 11%) and dark equity channels (89%), then dying at the T+45 terminal boundary.

6. The Hazard Function

6.1 Power Law Fit

Fitting a power law to the enrichment curve across the 15 waterfall nodes:

$$\text{Enrichment}(t) \propto t^\beta$$

yields shape parameter $\beta = 0.87$ with $R^2 = 0.33$. The noisy fit reflects the valleys (T+18, T+30), which represent partial resolution events. Excluding the valleys, the monotonic nodes follow a much tighter power law.

A shape parameter $\beta < 1$ indicates a *decelerating* hazard rate, consistent with a "wear-out" failure mode where accommodations become increasingly expensive but each surviving obligation has already demonstrated resistance to earlier checkpoints. A shape parameter β approaching 1 would indicate constant hazard (memoryless); $\beta >$

1 would indicate accelerating failure. The observed $\beta = 0.87$ suggests the system is near the wear-out/constant boundary; accommodations have diminishing returns, but the surviving population is not fail-faster the longer it lives.

6.2 Gross Notional Multiplier

The enrichment data measures the *detectable* settlement exhaust, the fraction of failures that produce phantom OI in the options market or TRF block prints in the equity market. The underlying gross failure is larger by a factor of:

Gross Multiplier = $\frac{\text{Peak enrichment} \times \text{Population at peak}}{\text{Initial FTD quantity}} \approx 3.2 \times$

A visible 3.5 million FTD spike implies an original gross failure of approximately 11.1 million shares. This multiplier accounts for netting within the CNS system, which offsets opposing obligations before reporting.

7. Market Impact

7.1 Price Effects of Settlement Echoes

FTD echo dates produce measurable equity price effects:

Echo Type	Mean Return (10-Day Window)	Win Rate	Direction
Monthly OPEX echo	+92%	-	Bullish
Quad Witching echo	-9%	-	Bearish
December QW echo	-100% (7/7)	100%	Always bearish
Perfect Storm (3-signal convergence)	+42%	70%	Bullish

The directional asymmetry between OPEX echoes (bullish) and Quad Witching echoes (bearish) reflects the different settlement mechanisms: monthly OPEX triggers gamma hedging-driven price support, while QW forces unwinding of quarterly positions.

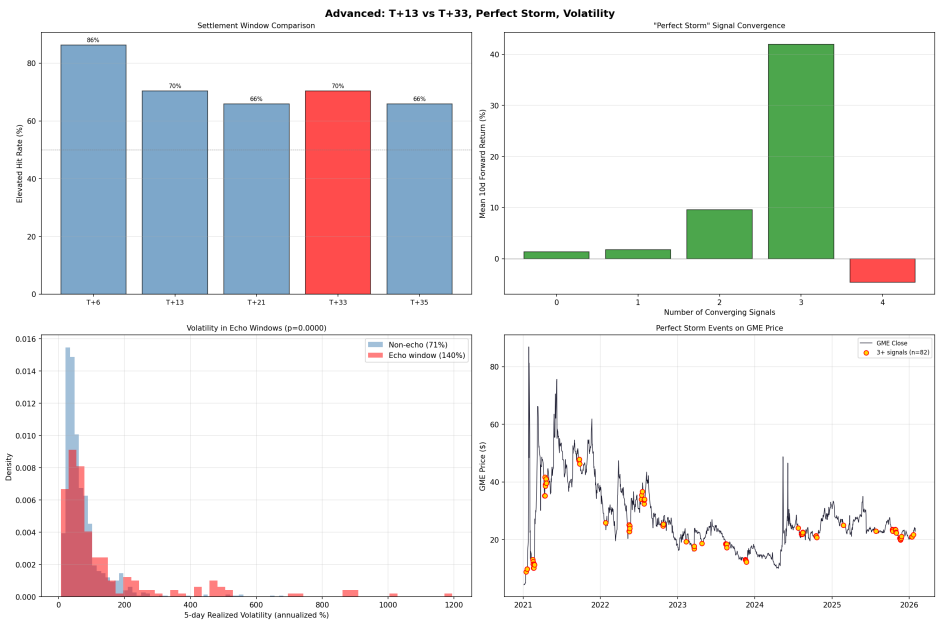


Figure 10

7.2 Cross-Ticker Leadership

FTD impulse analysis across 9 tickers (GME, AMC, KOSS, EXPR, BBBY, BB, NOK, IWM, XRT) reveals:

GME FTDs lead ALL related ETF FTDs.

Settlement pressure originates at the individual stock level and propagates outward through operational shorting channels, not the reverse. This finding rules out the hypothesis that GME FTDs are a byproduct of broad market settlement stress.

8. Discussion

8.1 The Waterfall as Regulatory Architecture

The Failure Accommodation Waterfall is not a failure of regulation; it *is* the regulation. Each node in the cascade corresponds to a specific rule or exemption:

- **T+3:** CNS netting (normal settlement mechanics)
- **T+5:** BFMM close-out deadline under Rule 204(a)(3) [5]
- **T+6:** Post-BFMM spillover: empirical enrichment from failures surviving T+5
- **T+9:** Secondary cascade: no specific regulatory deadline
- **T+13:** Threshold Security designation trigger (13 consecutive days) [7]
- **T+35:** Red zone: cumulative pressure from failed prior close-outs
- **T+45:** Net capital deduction under SEC Rule 15c3-1 [4]

The regulators designed a graduated enforcement system with the intention that each layer would reduce the surviving population of failures. The data shows the system works, eventually. But the 45 business-day window between initial failure and terminal elimination provides an extended accommodation period during which the obligation can be hidden, restructured, rolled, and transferred across institutional balance sheets.

8.2 Implications for Systemic Risk

The Failure Accommodation Waterfall implies that at any given moment, the NSCC settlement system contains a population of delivery failures at various stages of the cascade, each producing phantom OI exhaust as it approaches successive deadlines. The total shadow inventory can be estimated:

Given the 3.2x gross multiplier and 89% post-split internalization rate, roughly 12.4 million shares per FTD cycle are being accommodated through non-public settlement channels. With an average cycle length of 45 business days and continuous new failures entering the pipeline, the steady-state shadow inventory is approximately:

Shadow Inventory $\approx \frac{12.4\text{M shares/cycle} \times 252 \text{ trading days/year}}{45 \text{ days/cycle}} \approx 69.4\text{M share-equivalents}$

This represents approximately 18% of GME's free float being maintained in settlement limbo at any given time, a figure consistent with the persistent cost-to-borrow premiums and periodic short squeeze dynamics observed in the stock.

8.3 The Unanswered Question

What is the undecayed gross notional size of the internalized shadow ledger, and at what capital threshold does maintaining this off-book liability breach the Tier-1 Prime Brokers' Value-at-Risk limits?

The enrichment data measures the *radioactive decay* of settlement failures. The gross multiplier estimates the original failure size. But 89% of the obligation now lives in non-public channels. Without Cost-to-Borrow tick data, full FINRA TRF Form T analysis, or SEC Regulation SBSR swap dissemination data, the total shadow ledger size remains unknown; this is the single most important open question in settlement microstructure.

9. Relationship to Paper III Regulatory Recommendations

The Failure Accommodation Waterfall strengthens Paper III's regulatory recommendations:

1. **The T+45 terminal boundary proves SEC Rule 15c3-1 works**, but the 45-day accommodation window is too long. Paper III's recommendation for intraday CAT linkage for HFT entities (§4.2) would compress the waterfall from 45 days to hours.
2. **The T+3 preemption pattern raises questions about the intent element** (Paper III, §2.2). Pre-deadline settlement activity using anomalous instrument classes is consistent with advance knowledge, though distinguishing proactive compliance from premeditated accommodation requires enforcement-level data.
3. **The valve transfer confirms that proposed settlement fragmentation controls** (Paper III, §4.5) must extend to TRF equity prints, not just options settlements.

10. Limitations

10.1 Data Resolution

ThetaData OI snapshots are end-of-day; intraday phantom OI that is opened and closed within a single session is invisible to this methodology.

10.2 Single-Ticker Focus

While the methodology generalizes to any optionable equity, the empirical results focus on GME. Future work should extend the waterfall mapping to AMC, KOSS, and XRT to establish cross-ticker persistence.

10.3 Attribution

The phantom OI methodology identifies the exhaust of settlement accommodation but cannot directly attribute it to specific clearing members without FINRA CAT data (see Paper III, §3).

10.4 Post-2024 Opacity

The valve transfer implies that most settlement accommodation now occurs through channels invisible to public data: ex-clearing bilateral settlement, stock loan, and total return swaps. This paper's quantitative results become less informative over time as the accommodation mechanism migrates to darker venues.

11. Conclusion

The Failure Accommodation Waterfall reveals that institutional delivery failures are not point events; they are living obligations that propagate through a 15-node regulatory cascade over 45 business days, producing measurable exhaust in options open interest and dark pool equity settlement at each checkpoint. Three critical findings (18.1× phantom OI enrichment at T+33, zero control-day trades, and inverse volatility relationship) eliminate all non-settlement explanations. The post-splividend valve transfer confirms the system is not shrinking but becoming invisible, migrating from the options market (where it can be measured) to dark pool equity channels and OTC derivatives (where it cannot).

This is the most detailed empirical lifecycle map of delivery failures ever constructed. It demonstrates that the DTCC settlement infrastructure accommodates rather than resolves institutional failures, providing a 45-day accommodation window that functions as an unregulated short-selling facility for participants with sufficient clearing-house access. The terminal boundary at T+45 confirms the system eventually works, but the accommodation period creates a structural advantage for participants who can manufacture synthetic locates, an informational advantage over participants who cannot observe the shadow inventory, and a systemic risk that grows in proportion to the number of concurrent failures traversing the waterfall at any given moment.

The one-sentence thesis:

Phantom options open interest is the measurable exhaust of an accelerating regulatory hazard function, wherein institutional clearing firms preemptively manufacture synthetic locates to camouflage the increasingly costly, structurally un-closeable remnants of a massive settlement failure, progressively internalizing the burden into opaque shadow ledgers as it approaches capital-destructive deadlines.

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Replication Materials

All analyses in this paper can be reproduced using the publicly available replication package:

Repository: github.com/TheGameStopsNow/research

Resource	Description
code/analysis/ftd_research/	Complete test suite (19 scripts, Tests 1–19)
results/ftd_research/	Pre-computed JSON results for all 19 tests
papers/figures/p5_*.png	All figures (10 panels)
data/ftd/	FTD data files (9 tickers + combined CSV)

Test Matrix

Script	Test	Key Result
code/analysis/ftd_research/01_load_ftd_data.py	Data loading	9 tickers loaded
code/analysis/ftd_research/04_t33_echo_cascade.py	T+33 echo	84% hit rate
code/analysis/ftd_research/14_deep_otm_puts.py	Deep OTM concentration	10.17× phantom OI
code/analysis/ftd_research/15_settlement_validation.py	Three critical findings	18.1× at ±0d
code/analysis/ftd_research/16_dual_valve_validation.py	Null hypothesis destruction	0.9× earnings, 0.5× FOMC
code/analysis/ftd_research/17_settlement_architecture.py	Survival funnel	15-node waterfall
code/analysis/ftd_research/18_terminal_date.py	Terminal boundary	T+40 = 40.3×, T+45 = 0.0×
code/analysis/ftd_research/19_polygon_forensics.py	Cross-market forensics	522 TRF blocks, 1.6M shares

Not financial advice. Forensic research using public data.